

Numerical Investigation of Flow Losses in a 2D Tesla Valve

Francesco Trevisi, Dante Serrecchia, and Pietro Corsani Pio

Abstract—The work presented in this assignment aims to investigate the flow behaviour of a 2D Tesla valve. The numerical analysis focuses on pressure drops, velocity field and turbulence modelling. Yielded results are then compared both qualitatively and quantitatively with the turbulent Tesla valve benchmark work by Park and Kim.

I. INTRODUCTION

TESLA valves are passive flow-control devices characterized by a fixed internal geometry and by the absence of moving parts. Their operating principle relies on producing different hydraulic resistance depending on the flow direction. In the forward direction, the flow mainly follows the straight channel and the pressure loss is mostly associated with wall friction. In the reverse direction, part of the flow is diverted into the side branch and later rejoins the main stream, producing flow splitting, jet impingement, recirculation, stronger mixing and additional local losses. This directional behaviour is commonly quantified through the diodicity, defined as the ratio between the reverse and forward pressure drops. The present work focuses specifically on the reverse-flow configuration, where the largest pressure losses are expected and where the dominant loss mechanisms of the valve are most clearly visible.

Previous studies have shown that reverse-flow losses are strongly affected by the valve geometry, Reynolds number and flow structures near the diverging and merging regions. In particular, Park and Kim investigated the pressure-drop characteristics of Tesla valves in fully turbulent flow and analysed velocity, pressure and turbulent kinetic energy fields as indicators of the main loss mechanisms [1]. Their work provides a useful benchmark for assessing whether numerical simulations can reproduce the key features of flow splitting, jet impingement, turbulent mixing and pressure recovery in a Tesla-valve geometry.

In this work, a simplified two-dimensional Tesla valve is studied numerically using ANSYS Fluent. The objective is to compare the predicted reverse-flow pressure drop, streamwise and transverse velocity profiles, turbulent kinetic energy and pressure/velocity contours with the benchmark trends reported by Park and Kim. Since the reference configuration is three-dimensional, the comparison is not intended as a fully equivalent reproduction of the original valve, but as an assessment

of whether the main in-plane loss mechanisms can still be identified in the present simplified 2D model.

II. GEOMETRY AND NUMERICAL METHODS OLD

The objective of the numerical investigation consists of a two-dimensional Tesla valve unit, composed of a main straight channel and a curved side branch. The geometry was reconstructed from the benchmark paper into a 2D model to enable an affordable analysis.

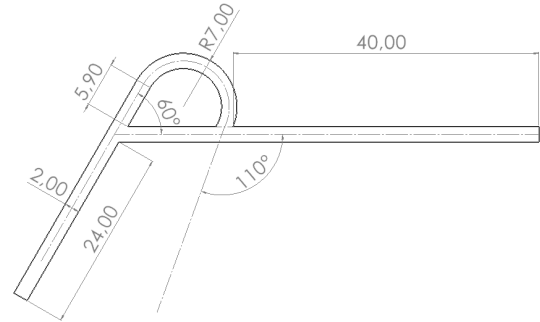


Figure 1. Two-dimensional Tesla valve geometry reconstructed from the benchmark configuration

The investigated configuration consists of a two-dimensional Tesla-valve unit composed of a straight main channel and an inclined side branch. Both channels have a constant width of 2 mm. The side branch departs from the main channel at a diverging angle of 110° , follows a 24 mm straight section and reconnects to the main channel through a curved path with radius $R = 7$ mm and a converging angle of 90° . A 40 mm downstream straight section was included after the merging region to allow the flow to redevelop before reaching the outlet.

We expect that going from a three-dimensional setup to a two-dimensional one will not resolve completely out-of-plane and secondary flow: as a result, the present model is a simplified representation intended to investigate in-plane flow separation phenomena, as well as reattachment and merging mechanisms.

III. GEOMETRY AND NUMERICAL METHODS

The numerical investigation considers a simplified two-dimensional Tesla-valve unit composed of a straight main channel and a curved side branch. The geometry was reconstructed from the benchmark configuration of Park and

F. Trevisi, D. Serrecchia, and P. Corsani Pio are with Delft University of Technology, Faculty of Mechanical Engineering, Mechanical Engineering - EFPT.

Project Group: 3. Report date: June 2026.

Kim [1] in order to obtain an affordable 2D model while preserving the main in-plane features responsible for reverse-flow losses.

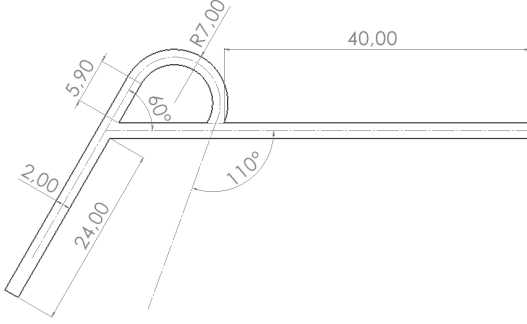


Figure 2. Two-dimensional Tesla valve geometry reconstructed from the benchmark configuration.

The investigated configuration consists of a straight main channel and an inclined side branch, both with a constant width of 2 mm. The side branch departs from the main channel at a diverging angle of 60° , follows a straight section, and reconnects to the main channel through a curved path with a converging angle of 110° . The branch curvature was chosen consistently with the benchmark reference geometry. A downstream straight section was included after the merging region to allow the flow to redevelop before reaching the outlet.

Reducing the original three-dimensional square-duct configuration to a two-dimensional model prevents the simulation from resolving out-of-plane and secondary-flow structures. Therefore, the present model is intended to investigate the dominant in-plane mechanisms of the reverse-flow case, including flow separation, reattachment, jet impingement and mixing at the merging region.

A. Governing Equations and Turbulence Closure

The simulation assumed steady, incompressible and isothermal flow. The mean flow field is governed by the Reynolds-averaged continuity and momentum equations:

$$\frac{\partial \bar{u}_i}{\partial x_i} = 0 \quad (1)$$

$$\rho \bar{u}_j \frac{\partial \bar{u}_i}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j} \left[\mu_{eff} \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) \right] - \frac{\partial}{\partial x_j} \left(\frac{2}{3} \rho k \delta_{ij} \right) \quad (2)$$

where $\bar{u}_i$ and $\bar{p}$ denote the Reynolds-averaged velocity and pressure fields, respectively. The term $-\rho \bar{u}_i' \bar{u}_j'$ represents the Reynolds stresses associated with the unresolved turbulent velocity fluctuations. Whereas, $\mu_{eff} = \mu + \mu_t$ includes the turbulent viscosity μ_t .

The Reynolds stresses were closed through the Boussinesq eddy-viscosity hypothesis:

$$-\rho \bar{u}_i' \bar{u}_j' = \mu_t \left(\frac{\partial \bar{u}_i}{\partial x_j} + \frac{\partial \bar{u}_j}{\partial x_i} \right) - \frac{2}{3} \rho k \delta_{ij} \quad (3)$$

where k is the turbulent kinetic energy.

Numerically, two different turbulence closures were implemented and compared in the simulation process, in order to follow the same procedure reported in the reference paper: both models solve transport equation for the turbulent kinetic energy k plus a second variable, used to determine the turbulent viscosity.

- Realizable $k - \epsilon$ model uses the dissipation energy ϵ as the second variable and it is usually robust and computationally efficient for fully turbulent internal flows.
- SST $k - \omega$ model uses specific dissipation rate ω , providing a more detailed treatment of the near-wall region and of adverse-pressure-gradient or separation phenomena.

Comparing their predictions of pressure drop, velocity fields and turbulent kinetic energy allows us to assess the sensitivity of the results to turbulence modelling assumptions and to determine which model better reproduces the benchmark trends reported in the reference paper.

B. Boundary Conditions and Operating Conditions

A uniform inlet velocity profile was imposed for each simulated Reynolds number. Reynolds number was evaluated as:

$$Re = \frac{\rho U_{in} D_h}{\mu} \quad (4)$$

where D_h is the hydraulic diameter of the inlet section. Since the present model is two-dimensional, the inlet channel width was used as the characteristic hydraulic length in a consistent manner throughout the study.

At the outlet, a zero gauge static pressure was prescribed, fixing the pressure reference level and allowing the velocity field to adjust according to the solution in the domain.

The valve walls were modelled as smooth, stationary and impermeable boundaries and a no-slip condition was applied, such as that:

$$u = v = 0 \quad (5)$$

The pressure drop across the valve was evaluated as the area-averaged static pressure difference between the selected upstream and downstream monitoring sections:

$$\Delta p = \langle p \rangle_{upstream} - \langle p \rangle_{downstream} \quad (6)$$

C. Numerical Solution Procedure

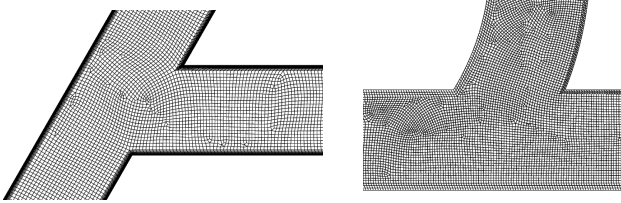
The simulations were performed in ANSYS Fluent using a pressure-based, steady-state solver. Pressure-velocity coupling was handled using the coupled algorithm. Second-order spatial discretization schemes were applied to the momentum and turbulence transport equations in order to limit numerical diffusion in the high-gradient regions near the diverging and merging sections.

Convergence was evaluated by monitoring both the scaled residuals and relevant integral quantities, particularly the pressure drop across the valve. The simulations were considered

converged once the residuals stabilized below the prescribed threshold of 10^{-6} and the monitored pressure drop showed no appreciable variation with additional iterations.

D. Meshing

The computational mesh was locally refined near the walls, in the curved side branch, and around the diverging and merging regions, where separation, reattachment, jet interaction and large pressure gradients are expected. Since the realizable $k-\epsilon$ and SST $k-\omega$ models rely on different near-wall treatments, separate mesh families were generated for the two turbulence closures.



(a) SST $k-\omega$ fine mesh (b) Realizable $k-\epsilon$ fine mesh

Figure 3. Local mesh refinement in the regions of strongest expected gradients. The SST $k-\omega$ mesh uses a finer near-wall resolution to resolve the viscous sublayer, while the realizable $k-\epsilon$ mesh is generated for wall-function treatment.

For the realizable $k-\epsilon$ model, wall functions were used. The first cell centre was therefore targeted in the logarithmic region of the turbulent boundary layer, with $y^+ \approx 30$. Three inflation layers were used with a growth rate of 1.1. For the SST $k-\omega$ model, the near-wall region was resolved directly, requiring the first cell centre to be located inside the viscous sublayer, with $y^+ \approx 1$. Fifteen inflation layers were therefore applied with a growth rate of 1.2.

This wall-treatment strategy follows the benchmark setup of Park and Kim, where the realizable $k-\epsilon$ model was combined with wall-function resolution and the SST $k-\omega$ model with low- y^+ near-wall resolution [1]. It is also consistent with the ANSYS Fluent near-wall modelling recommendations for wall-function and low-Reynolds-number treatments [2].

E. Mesh Sensitivity

To evaluate the influence of spatial discretization on the numerical solution, different grid resolutions were generated for the two turbulence models. Separate mesh families were used because the realizable $k-\epsilon$ and SST $k-\omega$ models require different near-wall resolutions. The characteristic grid size h_g denotes the reference element size in the core flow region, while N_c is the total number of computational cells. The main mesh parameters are summarized in Table I.

The mesh sensitivity was assessed by comparing the pressure drop and the velocity and turbulent kinetic energy profiles obtained on the different grid resolutions. For the realizable $k-\epsilon$ model, coarse, medium and fine meshes were explicitly compared. For the SST $k-\omega$ model, the final comparison was performed using the fine low- y^+ mesh, which contains a substantially larger number of cells because of the resolved near-wall treatment. Across the tested mesh range, the main

Table I
GRID RESOLUTIONS USED FOR THE REALIZABLE $k-\epsilon$ AND SST $k-\omega$ TURBULENCE MODELS.

Parameter	Turbulence model	Grid resolution		
		Coarse	Medium	Fine
h_g (mm)	Realizable $k-\epsilon$	0.20	0.14	0.10
	SST $k-\omega$	0.15	0.10	0.07
N_c	Realizable $k-\epsilon$	15249	29435	55843
	SST $k-\omega$	—	—	163704
Target y^+	Realizable $k-\epsilon$	≈ 30	≈ 30	≈ 30
	SST $k-\omega$	≈ 1	≈ 1	≈ 1
Max. skewness	Realizable $k-\epsilon$	0.752	0.765	0.775
	SST $k-\omega$	—	—	—
Min. orthogonal quality	Realizable $k-\epsilon$	0.397	0.381	0.369
	SST $k-\omega$	—	—	0.344

flow structures and the pressure-drop trend remained consistent. The largest differences were observed near the merging region, where velocity gradients and turbulent kinetic energy production are strongest. For this reason, the fine meshes were selected for the final benchmark comparison.

IV. RESULTS AND BENCHMARK COMPARISON

The numerical results are compared with the benchmark data of Park and Kim [1] to assess whether the simplified two-dimensional model reproduces the main reverse-flow mechanisms of the reference Tesla-valve configuration. The comparison combines qualitative and quantitative information. Velocity and turbulent kinetic energy profiles are used to identify the dominant flow structures along the main channel, while the pressure-drop results are used to evaluate the reverse-flow hydraulic losses predicted by the simulations.

A. Mean Velocity and Turbulent Kinetic Energy Profiles

The mean streamwise velocity $\bar{u}_x$, the mean transverse velocity $\bar{u}_y$, and the turbulent kinetic energy k were extracted along the main channel and compared with digitized profiles from the benchmark study. The comparison was performed for the realizable $k-\epsilon$ and SST $k-\omega$ turbulence models, including the available mesh resolutions for each model. In the plots, the turbulence closures are distinguished by colour, while the mesh resolution is distinguished by line style.

1) **Streamwise Velocity Profile:** Figure 4 shows the comparison of the mean streamwise velocity component, $\bar{u}_x$, along the main channel.

All numerical simulations reproduce the main trend of the benchmark profile. In the upstream part of the channel, the streamwise velocity remains relatively low. A sharp acceleration is then observed near the merging region, approximately between $x = 13$ mm and $x = 15$ mm. This acceleration is associated with the local restriction of the effective flow passage and with the rejoining of the side-branch flow into the main channel.

Downstream of the merging region, the Fluent simulations predict an almost constant streamwise velocity of approximately 25.5–26 m/s. The benchmark profile shows a similar abrupt acceleration, but it reaches slightly higher

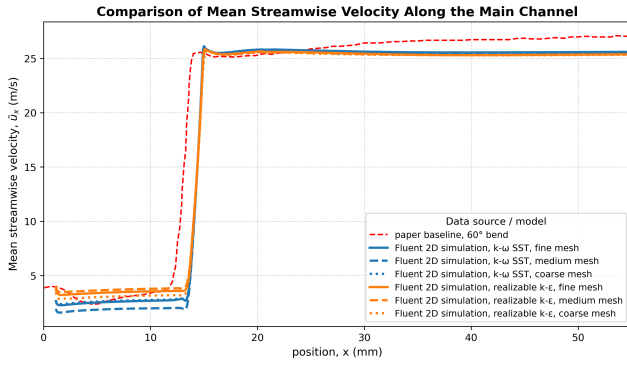


Figure 4. Comparison of the mean streamwise velocity profile $\bar{u}_x$ along the main channel for the digitized benchmark data and the Fluent simulations using the SST $k - \omega$ and realizable $k - \epsilon$ turbulence models.

values farther downstream, approaching approximately 27 m/s. This indicates that the simulations capture the location and shape of the acceleration region well, but slightly underpredict the downstream streamwise velocity recovery.

The SST $k - \omega$ and realizable $k - \epsilon$ models produce very similar streamwise velocity profiles after the merging region. The difference between the two turbulence models is more visible upstream of the acceleration region, where the realizable $k - \epsilon$ simulations predict slightly larger streamwise velocities. Regarding mesh sensitivity, the fine, medium and coarse meshes provide nearly overlapping results, especially downstream of the merging region. This suggests that the streamwise velocity prediction is only weakly dependent on the grid resolution for the present mesh range.

2) Transverse Velocity Profile: Figure 5 compares the mean transverse velocity component, $\bar{u}_y$, along the main channel.

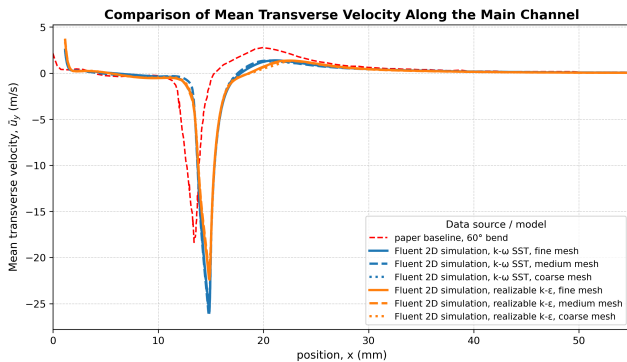


Figure 5. Comparison of the mean transverse velocity profile $\bar{u}_y$ along the main channel for the digitized benchmark data and the Fluent simulations using the SST $k - \omega$ and realizable $k - \epsilon$ turbulence models.

The transverse velocity profile highlights the strong flow deflection occurring in the merging region. Upstream of this region, the transverse velocity remains close to zero, showing that the flow is mainly aligned with the main channel. Around $x = 13$ – 16 mm, all profiles exhibit a strong negative peak. This minimum corresponds to the downward deflection of the flow caused by the interaction between the side-branch

jet and the main-channel flow.

The numerical simulations predict a more intense negative transverse velocity than the benchmark data. The Fluent profiles reach values of approximately -22 to -26 m/s, while the benchmark profile reaches approximately -18 to -19 m/s. This indicates that the simplified two-dimensional simulations predict a stronger local flow deflection in the merging region.

After the negative peak, the transverse velocity changes sign and reaches a positive maximum around $x = 20$ – 23 mm. This behaviour represents the recovery and reorientation of the flow after the jet interaction region. Farther downstream, all profiles decay toward zero, indicating that the flow gradually realigns with the main-channel direction.

The SST $k - \omega$ model predicts a slightly stronger negative peak than the realizable $k - \epsilon$ model. This is consistent with the stronger sensitivity of the SST $k - \omega$ model to near-wall and separated-flow effects. The fine and medium meshes are very close to each other, while the coarse mesh shows small deviations near the region of maximum gradient. Therefore, the transverse velocity profile is more sensitive to both turbulence model and mesh resolution than the streamwise velocity profile.

3) Turbulent Kinetic Energy Profile: Figure 6 shows the turbulent kinetic energy profile, k , along the main channel.

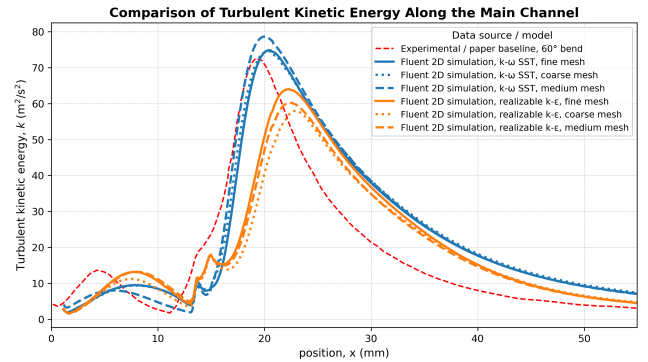


Figure 6. Comparison of the turbulent kinetic energy profile k along the main channel for the digitized benchmark data and the Fluent simulations using the SST $k - \omega$ and realizable $k - \epsilon$ turbulence models.

The turbulent kinetic energy profile confirms that the strongest turbulence production occurs near the merging region. In the upstream portion of the channel, the values of k remain relatively small. A first moderate increase is observed before the main interaction region, followed by a much stronger peak around $x = 19$ – 23 mm. This peak corresponds to the region where the branch flow rejoins the main stream, producing strong shear, mixing and turbulent energy generation.

The SST $k - \omega$ model predicts the largest turbulent kinetic energy levels, with a peak of approximately 75 – 79 m^2/s^2 . The benchmark profile reaches a slightly lower peak of

approximately $72\text{--}73 \text{ m}^2/\text{s}^2$, while the realizable $k - \epsilon$ model predicts lower peak values, approximately $58\text{--}64 \text{ m}^2/\text{s}^2$. Therefore, the SST $k - \omega$ model gives a closer prediction of the peak magnitude of turbulent kinetic energy, although it slightly delays the location of the maximum compared with the benchmark curve.

The realizable $k - \epsilon$ model underpredicts the maximum turbulent kinetic energy in the main mixing region. This suggests that this model is more dissipative for the present flow configuration and may smooth the strong shear-layer interaction occurring after the side branch rejoins the main channel. However, farther downstream, both turbulence models predict the gradual decay of turbulent kinetic energy, which is consistent with the progressive recovery of the flow.

The mesh sensitivity is relatively small for both turbulence models. The fine, medium and coarse meshes follow the same overall trend and produce similar downstream decay. The largest mesh-related differences occur near the turbulent kinetic energy peak, where the gradients are strongest. This confirms that the merging region is the most critical part of the domain for grid refinement.

4) Qualitative comparison of pressure and velocity contours. The normalized pressure and velocity contours obtained on the fine grid reproduce the main flow features reported by Park and Kim for the reverse-flow configuration. In the benchmark contours, the normalized quantities are plotted approximately in the range $[-0.60, 1.50]$, and the most relevant flow structures are located around the branch-main-channel junction. Similarly, in the present simulations, both turbulence models predict a high-pressure region upstream of the merging point and inside the curved branch, followed by a sharp pressure decrease immediately downstream of the junction. This confirms that the dominant pressure loss is generated by the recombination of the branch flow with the main-channel flow, which produces jet impingement, strong local acceleration and enhanced mixing.

The velocity contours also show good qualitative agreement with the reference results. In all cases, a high-speed jet develops along the lower part of the outlet channel after the merging point, while a low-velocity region forms close to the upper wall. In the benchmark, this low-velocity zone is associated with a clearly visible recirculation region, as shown by the streamlines below the velocity contour. The same mechanism is captured in the present simulations, although the separated region appears smoother and less sharply defined, especially for the realizable $k - \epsilon$ model. This difference can be related to the simplified two-dimensional geometry used in this work, which cannot reproduce the out-of-plane secondary motions present in the three-dimensional square duct considered in the reference study. Moreover, the contour ranges are not identical: the present pressure fields reach wider normalized ranges, particularly for the SST $k - \omega$ model, while the velocity contours are plotted from zero to the local maximum value. Therefore, the colour distribution should not be compared directly, but rather used to identify

the main flow structures.

Comparing the two turbulence closures, the SST $k - \omega$ model predicts stronger local gradients near the merging region. This is visible from the deeper low-pressure pocket downstream of the junction and from the slightly higher maximum normalized velocity in the outlet jet. The high-speed region is more localized than in the realizable $k - \epsilon$ case, indicating a stronger prediction of the jet-impingement and separation effects. This behaviour is consistent with the near-wall treatment of the SST $k - \omega$ model, which resolves the viscous sublayer with a target $y^+ \approx 1$ and is generally more sensitive to adverse pressure gradients and separated flows. Conversely, the realizable $k - \epsilon$ model, used with wall functions and a target $y^+ \approx 30$, produces a smoother and more diffused flow field. Overall, both models capture the same loss mechanism observed in the benchmark, but the SST $k - \omega$ solution emphasizes the local pressure and velocity gradients more strongly, while the realizable $k - \epsilon$ solution provides a more averaged prediction of the recombination region.

B. Pressure-drop parametric study

Figure 6 compares the reverse-flow pressure drop obtained from the Fluent simulations with the benchmark curves for both turbulence models over the investigated Reynolds-number range.

For both turbulence closures, the pressure drop increases nonlinearly with Reynolds number. This trend is physically expected because the dynamic losses associated with flow splitting, branch-channel curvature, jet impingement and turbulent mixing become progressively stronger as the flow rate increases.

The SST $k - \omega$ simulation provides the closest agreement with the corresponding benchmark curve. At $Re = 50,000$, the present SST result is approximately $6.9 \times 10^5 \text{ Pa}$, compared with about $7.1 \times 10^5 \text{ Pa}$ in the benchmark. The relative deviation is therefore limited to roughly 3%.

The realizable $k - \epsilon$ model predicts systematically lower pressure drops than the benchmark realizable $k - \epsilon$ trend. At $Re = 50,000$, the present result is approximately $5.85 \times 10^5 \text{ Pa}$, while the benchmark value is around $6.3 \times 10^5 \text{ Pa}$, corresponding to an underprediction of approximately 7%.

The discrepancy becomes more evident as Reynolds number increases. This suggests that the simplified 2D configuration does not reproduce all loss-generating mechanisms present in the three-dimensional reference geometry. In particular, out-of-plane motion and secondary-flow structures may contribute to additional momentum redistribution and dissipation in the physical valve.

Despite these differences, both turbulence models reproduce the correct monotonic and nonlinear dependence of pressure drop on Reynolds number. The SST $k - \omega$ model shows better

agreement with the pressure-drop benchmark, whereas the realizable $k-\epsilon$ model provides a slightly lower-loss prediction over the full Reynolds-number range.

C. Pressure-Drop Reynolds-Number Study

Figure 9 compares the reverse-flow pressure drop obtained from the Fluent simulations with the digitized benchmark data of Park and Kim [1] over the investigated Reynolds-number range.

For both turbulence closures, the pressure drop increases nonlinearly with Reynolds number. This behaviour is physically expected because increasing flow rate strengthens the dynamic losses associated with flow splitting, branch-channel curvature, jet impingement and turbulent mixing.

The SST $k-\omega$ simulation provides the closest agreement with the corresponding benchmark trend. At $Re = 50,000$, the present SST result is approximately 6.9×10^5 Pa, compared with about 7.1×10^5 Pa in the benchmark, corresponding to a relative deviation of roughly 3%.

The realizable $k-\epsilon$ model predicts systematically lower pressure drops than the corresponding benchmark trend. At $Re = 50,000$, the present result is approximately 5.85×10^5 Pa, while the benchmark value is around 6.3×10^5 Pa, corresponding to an underprediction of approximately 7%.

The discrepancy becomes more pronounced as Reynolds number increases, suggesting that the simplified 2D configuration does not reproduce all loss-generating mechanisms present in the three-dimensional reference geometry. In particular, out-of-plane motion and secondary-flow structures may contribute to additional momentum redistribution and dissipation in the physical valve.

Despite these limitations, both turbulence models reproduce the correct monotonic and nonlinear dependence of reverse-flow pressure drop on Reynolds number. The SST $k-\omega$ model gives the closest pressure-drop prediction in the present comparison, whereas the realizable $k-\epsilon$ model provides a lower-loss estimate over the investigated range.

D. Benchmark comparison and limitations

This study investigates the flow behaviour of a simplified two-dimensional Tesla valve using ANSYS Fluent. The analysis focuses on the backward-flow configuration, where the interaction between the side-branch jet and the main-channel flow produces flow deflection, reattachment, mixing and additional pressure losses.

The simplified two-dimensional geometry allows the main in-plane mechanisms to be studied at reduced computational cost. However, it cannot capture out-of-plane motion or secondary-flow effects, which may influence the pressure-drop characteristics of a real three-dimensional Tesla valve. Therefore, the results should be interpreted as a simplified comparison with the benchmark study by Park and Kim rather than as a fully equivalent reproduction.

E. Benchmark Comparison and Limitations NEW

The present study investigates the reverse-flow behaviour of a simplified two-dimensional Tesla valve using ANSYS Fluent. The comparison with Park and Kim [1] shows that the main in-plane mechanisms of the benchmark configuration are reproduced: flow deflection at the merging region, acceleration of the outlet jet, enhanced turbulent kinetic energy production and a nonlinear increase of pressure drop with Reynolds number. However, the comparison should be interpreted with some limitations. First, the reference configuration is a three-dimensional square duct, whereas the present model is two-dimensional. As a result, out-of-plane motion and secondary-flow structures cannot be resolved, which may affect momentum redistribution, recirculation and pressure recovery. Second, the analysis focuses on the reverse-flow configuration only; therefore, the full diodicity of the valve is not evaluated. Finally, part of the comparison relies on digitized benchmark profiles and normalized contour fields, so the results are more

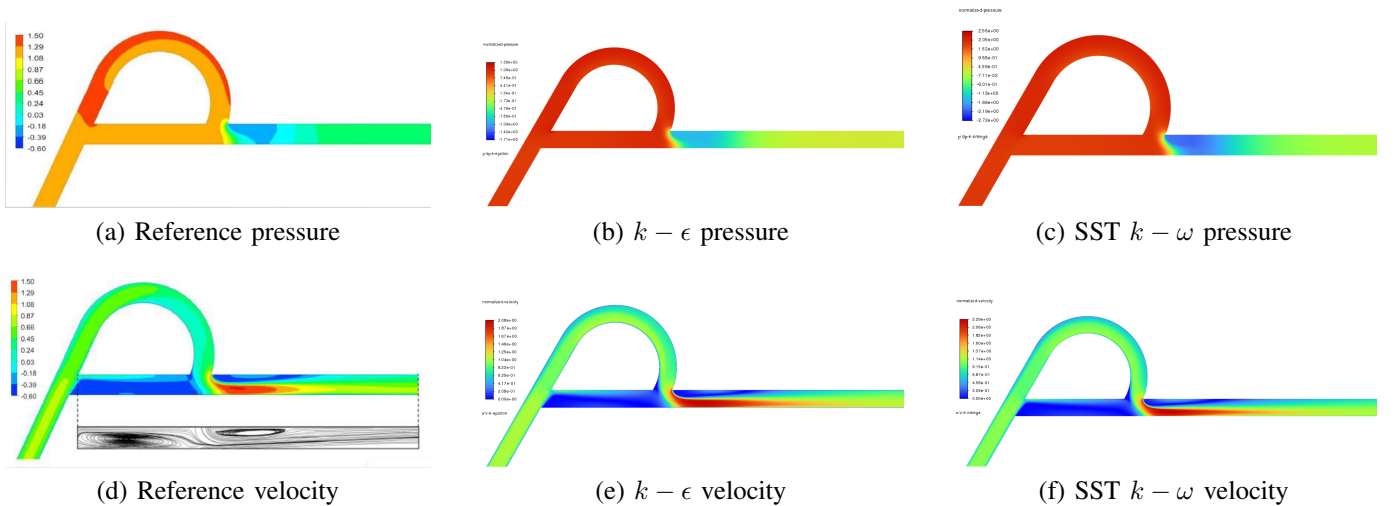


Figure 7. Comparison of pressure and velocity magnitude contours between the benchmark reference case and the fine-grid numerical simulations obtained with the realizable $k-\epsilon$ and SST $k-\omega$ turbulence models. The pressure contours show the pressure-loss distribution along the valve, while the velocity contours highlight the acceleration, flow redirection and interaction between the side-branch jet and the main-channel flow.

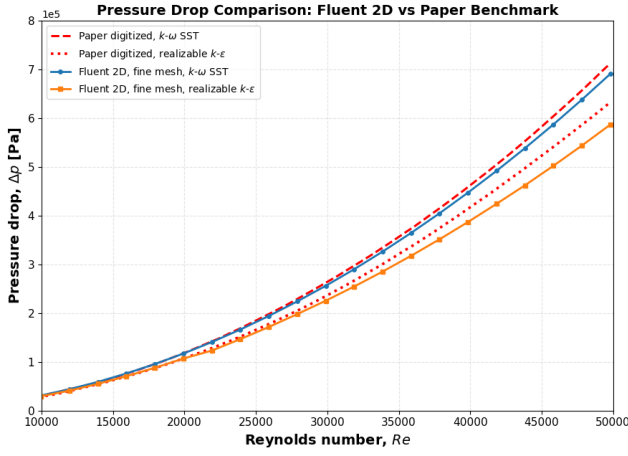


Figure 8. Reverse-flow pressure drop as a function of Reynolds number for the simplified 2D Tesla valve. The Fluent results obtained with the realizable $k-\epsilon$ and SST $k-\omega$ turbulence models are compared with the digitized benchmark data from Park and Kim [1]. Both models reproduce the nonlinear increase of pressure drop with Reynolds number, with the SST $k-\omega$ model showing the closest agreement with the reference trend.

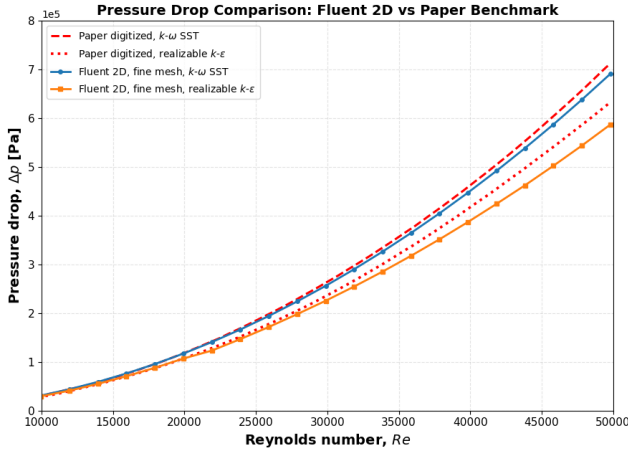


Figure 9. Reverse-flow pressure drop as a function of Reynolds number for the simplified 2D Tesla valve. The Fluent results obtained with the realizable $k-\epsilon$ and SST $k-\omega$ turbulence models are compared with the digitized benchmark data from Park and Kim [1]. Both models reproduce the nonlinear increase of pressure drop with Reynolds number, with the SST $k-\omega$ model showing the closest agreement with the reference trend in the investigated range.

suitable for assessing trends and dominant loss mechanisms than for exact pointwise validation.

Despite these limitations, the agreement in the velocity profiles, turbulent kinetic energy distribution and pressure-drop trend indicates that the simplified 2D model captures the dominant reverse-flow loss mechanisms of the Tesla-valve geometry.

1) Overall Discussion: Overall, the simulations reproduce the main mechanisms observed in the benchmark profiles: streamwise acceleration in the merging region, strong transverse deflection caused by the side-branch jet, and a turbulent kinetic energy peak associated with shear-layer interaction and mixing.

The streamwise velocity profile shows good agreement

with the benchmark data, especially in the location of the acceleration region. Larger differences are observed in the transverse velocity profile, where the simulations predict a stronger negative peak. This suggests that the two-dimensional model may overestimate the in-plane flow deflection because it cannot redistribute momentum through out-of-plane motion.

The turbulent kinetic energy results show a clear dependence on the turbulence model. The SST $k-\omega$ model predicts a peak closer to the benchmark magnitude, while the realizable $k-\epsilon$ model gives lower turbulent kinetic energy levels, indicating a more dissipative behaviour in the main mixing region.

APPENDIX A

ADDITIONAL GEOMETRY INFORMATION

Additional geometry details, dimensions or reconstruction assumptions can be included here.

APPENDIX B

MESH AND SOLVER DETAILS

Additional mesh-quality metrics, mesh-independence results, solver settings and residual convergence criteria can be included here.

REFERENCES

- [1] H. Park and S.-Y. Kim, "Pressure drop characteristics of tesla valve in fully turbulent flow," *Journal of Fluids Engineering*, vol. 148, no. 3, p. 031206, 2026.
- [2] ANSYS Inc., *ANSYS Fluent Theory Guide*, 2024, turbulence modelling theory guide.